

<!DOCTYPE html>

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<section class="cover">
<div>
  <p class="eyebrow">Revision 2026 Handbook</p>
  <h1>Engineering Drawing with CAD</h1>
  <p class="subtitle">A comprehensive bilingual laboratory guide covering geometrical construction</p>
  <div class="badges">
    <span class="badge">Course 1004</span>
    <span class="badge">REV2026</span>
    <span class="badge">1.5 Credits</span>
    <span class="badge">Bilingual (EN/ML)</span>
  </div>
  <div class="actions no-print">
    <a class="btn primary" href="#roadmap">Start Learning</a>
    <a class="btn ghost" href="/notes/downloadable-notes-1004.pdf">Download PDF</a>
  </div>
</div>
<aside class="course-facts">
  <h2>Course Overview</h2>
  <dl>
    <dt>Course Code</dt><dd>1004</dd>
    <dt>Semester</dt><dd>I</dd>
    <dt>Instructional Hours</dt><dd>45 Hours</dd>
    <dt>Internal (CIE)</dt><dd>60 Marks</dd>
    <dt>External (ESE)</dt><dd>40 Marks</dd>
    <dt>Course Outcomes</dt><dd>C01, C02, C03, C04</dd>
  </dl>
</aside>
</section>

<section class="section no-print" id="roadmap">
  <h2>Learning Roadmap</h2>
  <div class="roadmap">
    <a href="#intro">Introduction</a>
    <a href="#module-I">Geometrical & Dev.</a>
    <a href="#module-II">Projections & Sections</a>
    <a href="#module-III">CAD Basics</a>
    <a href="#module-IV">CAD Practice</a>
  </div>
</section>

<section class="section" id="intro">
  <h2>1. Introduction to Engineering Graphics</h2>
  <p class="lead">Engineering Graphics is the universal language of engineers. It allows for the precise communication of design ideas through technical drawings. This section introduces the fundamental principles and standards of engineering graphics, covering the use of projection, orthographic views, and descriptive geometry. It emphasizes the importance of accuracy and clarity in technical communication, which is essential for the design and manufacturing process. The section also introduces the standards governing the preparation of engineering drawings, ensuring consistency and interoperability across different engineering disciplines and organizations. The learning objectives of this section include understanding the purpose and importance of engineering graphics, identifying the various types of engineering drawings, and applying the standards and conventions for the preparation of technical drawings. The section concludes with a summary of the key concepts and a list of recommended resources for further study. The following table provides a detailed overview of the topics covered in this section, including the learning objectives, key concepts, and recommended resources. The table is organized into columns for the topic, learning objectives, key concepts, and recommended resources. The first column lists the topics to be covered, the second column lists the learning objectives for each topic, the third column lists the key concepts to be understood, and the fourth column lists the recommended resources for further study. The table is designed to provide a comprehensive overview of the section's content and to help students plan their learning activities. The table is as follows:
  <table>
    <thead>
      <tr>
        <th>Topic</th>
        <th>Learning Objectives</th>
        <th>Key Concepts</th>
        <th>Recommended Resources</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>Introduction to Engineering Graphics</td>
        <td>Understand the purpose and importance of engineering graphics. Identify the various types of engineering drawings.</td>
        <td>Technical communication, design process, types of drawings.</td>
        <td>Engineering Graphics Handbook, BIS SP 46, Engineering Drawing with CAD.</td>
      </tr>
      <tr>
        <td>Standards of Engineering Graphics</td>
        <td>Understand the importance of standards in engineering graphics. Identify the various standards governing the preparation of engineering drawings.</td>
        <td>BIS SP 46, Engineering Drawing with CAD, Engineering Graphics Handbook.</td>
        <td>Engineering Graphics Handbook, BIS SP 46, Engineering Drawing with CAD.</td>
      </tr>
      <tr>
        <td>Orthographic Projections</td>
        <td>Understand the principles of orthographic projection. Identify the various types of orthographic views.</td>
        <td>First angle projection, orthographic views, projection of points, lines, and planes.</td>
        <td>Engineering Graphics Handbook, BIS SP 46, Engineering Drawing with CAD.</td>
      </tr>
      <tr>
        <td>Descriptive Geometry</td>
        <td>Understand the principles of descriptive geometry. Identify the various types of descriptive geometry problems.</td>
        <td>Orthographic projection, descriptive geometry, construction of points, lines, and planes.</td>
        <td>Engineering Graphics Handbook, BIS SP 46, Engineering Drawing with CAD.</td>
      </tr>
    </tbody>
  </table>
  <div class="grid two">
    <article class="card">
      <h3>Drawing Standards (BIS SP 46)</h3>
      <p>All drawings must follow the Bureau of Indian Standards (BIS) code SP 46. This ensures uniformity and clarity in technical communication. The standards cover various aspects of drawing preparation, including line types, lettering, scales, and dimensions. Adherence to these standards is essential for the effective communication of design ideas through technical drawings. The following table provides a detailed overview of the standards covered in this section, including the standard number, the standard title, and the key requirements. The table is organized into columns for the standard number, the standard title, and the key requirements. The first column lists the standard numbers, the second column lists the standard titles, and the third column lists the key requirements. The table is designed to provide a comprehensive overview of the standards and to help students understand the requirements for the preparation of technical drawings. The table is as follows:
      <table>
        <thead>
          <tr>
            <th>Standard Number</th>
            <th>Standard Title</th>
            <th>Key Requirements</th>
          </tr>
        </thead>
        <tbody>
          <tr>
            <td>BIS SP 46</td>
            <td>Code of Practice for Drawing</td>
            <td>Line types, lettering, scales, dimensions, and other drawing conventions.</td>
          </tr>
          <tr>
            <td>BIS SP 47</td>
            <td>Code of Practice for Drawing</td>
            <td>Line types, lettering, scales, dimensions, and other drawing conventions.</td>
          </tr>
          <tr>
            <td>BIS SP 48</td>
            <td>Code of Practice for Drawing</td>
            <td>Line types, lettering, scales, dimensions, and other drawing conventions.</td>
          </tr>
        </tbody>
      </table>
    </article>
    <article class="card">
      <h3>Drawing Instruments</h3>
      <p>Professional manual drafting requires a set of precision tools. Even in the CAD era, understanding the use of drawing instruments is essential for the preparation of technical drawings. This section introduces the various types of drawing instruments, including the T-square, set squares, compass, and divider. It also discusses the proper use and maintenance of these instruments. The learning objectives of this section include identifying the various types of drawing instruments, understanding the proper use and maintenance of these instruments, and applying the instruments to the preparation of technical drawings. The section concludes with a summary of the key concepts and a list of recommended resources for further study. The following table provides a detailed overview of the instruments covered in this section, including the instrument name, the instrument description, and the recommended resources. The table is organized into columns for the instrument name, the instrument description, and the recommended resources. The first column lists the instrument names, the second column lists the instrument descriptions, and the third column lists the recommended resources. The table is designed to provide a comprehensive overview of the instruments and to help students understand the proper use and maintenance of these instruments. The table is as follows:
      <table>
        <thead>
          <tr>
            <th>Instrument Name</th>
            <th>Instrument Description</th>
            <th>Recommended Resources</th>
          </tr>
        </thead>
        <tbody>
          <tr>
            <td>T-square</td>
            <td>Used for drawing horizontal and vertical lines.</td>
            <td>Engineering Graphics Handbook, BIS SP 46, Engineering Drawing with CAD.</td>
          </tr>
          <tr>
            <td>Set Squares</td>
            <td>Used for drawing lines at specific angles (30°, 45°, 60°).</td>
            <td>Engineering Graphics Handbook, BIS SP 46, Engineering Drawing with CAD.</td>
          </tr>
          <tr>
            <td>Compass</td>
            <td>Used for drawing arcs and circles.</td>
            <td>Engineering Graphics Handbook, BIS SP 46, Engineering Drawing with CAD.</td>
          </tr>
          <tr>
            <td>Divider</td>
            <td>Used for dividing lines into equal parts and for drawing arcs and circles.</td>
            <td>Engineering Graphics Handbook, BIS SP 46, Engineering Drawing with CAD.</td>
          </tr>
        </tbody>
      </table>
    </article>
  </div>
  <div class="glossary-grid">
    <div>
      <dt>Mini Drafter</dt>
      <dd>Used for drawing parallel, perpendicular, and inclined lines</dd>
    </div>
    <div>
      <dt>Set Squares</dt>
      <dd>30°-60° and 45° squares for standard angles.</dd>
    </div>
    <div>
      <dt>Pencils</dt>
      <dd>H/2H for construction (light), HB/H for outlines (dark).</dd>
    </div>
  </div>
</section>
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    </div>
</article>
</div>

<div class="table-wrap">
  <table>
    <caption>Drawing Standards Checklist</caption>
    <thead><tr><th>Feature</th><th>Standard/Rule</th><th>Application</th></tr></thead>
    <tbody>
      <tr><td>Dimensioning</td><td>Aligned or Unidirectional system</td><td>Placing sizes on drawings</td></tr>
      <tr><td>Arrowheads</td><td>Length = 3 × Width (closed & filled)</td><td>Ending dimension lines</td></tr>
      <tr><td>Hatching</td><td>Thin lines at 45° to the axis</td><td>Representing cut surfaces</td></tr>
      <tr><td>Title Block</td><td>Bottom right corner (170mm x 65mm)</td><td>Drawing identification</td></tr>
    </tbody>
  </table>
</div>
</section>

<section class="module" id="module-I">
  <div class="module-head">
    <div>
      <p class="eyebrow">Module I</p>
      <h2>Geometrical Constructions and Developments</h2>
      <p class="lead">Construct regular polygons, exact conic sections, and visualize 3D surfaces </p>
    </div>
    <div class="module-meta">
      <span>9 Hours</span><span>C01</span><span>Apply</span>
    </div>
  </div>

  <div class="ml-note" lang="ml">
    <strong>Malayalam Support:</strong> വശങ്ങളുടെ നീളം നൽകിയാൽ പെന്റഗൺ, ഹെക്സഗൺ എന്നിവ നി
  </div>

  <section class="concepts">
    <h3>1.1 Regular Polygons</h3>
    <p>A regular polygon has all sides and internal angles equal. For Course 1004, you must master</p>
    <div class="topic-grid">
      <article class="topic-card">
        <h3>General Method</h3>
        <p>Can be used for any polygon. Draw the given side AB. At B, draw a perpendicular. With B
        </p>
      </article>
      <article class="topic-card">
        <h3>Angle Method</h3>
        <p>Internal angle of Pentagon = 108°. Internal angle of Hexagon = 120°. Use a protractor to
        </p>
      </article>
    </div>

    <h3>1.2 Conic Sections: The Ellipse</h3>
    <p>An ellipse is a curve where the sum of distances from two fixed points (foci) is constant. </p>
    <div class="grid two">
      <article class="worked">
        <h3>Rectangular (Oblong) Method</h3>
        <ol>
          <li>Draw a rectangle with sides equal to the Major and Minor axes.</li>
          <li>Divide the axes and the rectangle sides into equal segments (e.g., 4 parts).</li>
          <li>Project lines from the vertices of the minor axis through these points to find the c
          </li>
        </ol>
      </article>
      <article class="worked">
        <h3>Concentric Circle Method</h3>
        <ol>
          <li>Draw two circles with diameters equal to the Major and Minor axes.</li>
        </ol>
      </article>
    </div>
  </div>

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        <li>Divide circles into 12 equal sectors (30° each).</li>
        <li>From outer circle points, draw vertical lines; from inner circle points, draw horizontal lines.
    </ol>
</article>
</div>
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<h3>1.3 Development of Surfaces</h3>

<p>Development is the unfolding of a solid's surface onto a single plane. This is critical in understanding the geometry of 3D objects and their 2D representations.</p>

<div class="topic-grid">

<article class="topic-card">

<h3>Rectangular Prism</h3>

<p>The development consists of four rectangles (lateral faces) joined side-by-side, plus two rectangles for the top and bottom bases.</p>

</article>

<article class="topic-card">

<h3>Cylinder</h3>

<p>The development of the curved surface is a rectangle. Height = Cylinder Height and Width = Circumference.</p>

</article>

</div>

</section>

<section class="practical-centre">

<h3>Lab Exercises (Module I)</h3>

<article class="experiment" id="exp-i-1">

<header>Ex. I.1<div><h3>Polygon Construction</h3></div></header>

<div class="official-outcome">Outcome: Construct a regular pentagon and hexagon.</div>

<div class="exp-grid">

<section><h4>Procedure (Hexagon)</h4>Draw base AB = 40mm.With A and B as centres, draw arcs of radius 40mm intersecting at C.Repeat the process to find D, E, and F, completing the hexagon.</section>

<section class="safety"><h4>Pro Tip</h4><p>For a pentagon, use the 108° method for highest accuracy.</p></section>

</div>

</article>

<article class="experiment" id="exp-i-2">

<header>Ex. I.2<div><h3>Cylinder Development</h3></div></header>

<div class="official-outcome">Outcome: Draw the development of a cylinder (lateral surface only).</div>

<div class="exp-grid">

<section><h4>Procedure</h4>Calculate Circumference: $3.14 \times 50 = 157\text{mm}$.Draw a horizontal line of length 157mm, labeled CD.Divide CD into 12 equal parts, labeled P1 to P11 from C to D.Draw vertical lines from C, P1, P2, ..., P11, and D, extending upwards and downwards to a height of 50mm (the cylinder's height).Draw arcs centered at C and D with radius 50mm, intersecting the vertical lines at points Q1, Q2, ..., Q11 on the top and bottom edges respectively.Smoothly connect the points Q1, P1, P2, ..., P11, Q11 on the top edge and C, P1, P2, ..., P11, D on the bottom edge to form the curved surface of the cylinder's development.</section>

<section><h4>Calculation</h4><p> $C = \pi \times D$
 $C = 3.14159 \times 50 \approx 157.1\text{ mm}$ </p></section>

</div>

</article>

</section>

</section>

<section class="module" id="module-II">

<div class="module-head">

<div>

<p class="eyebrow">Module II</p>

<h2>Orthographic Projections and Sectional Views</h2>

<p class="lead">Master the theory of 2D representation of 3D objects and internal detailing through practical exercises.</p>

</div>

<div class="module-meta">

15 HoursC02Apply

</div>

</div>

<div class="ml-note" lang="ml">

Malayalam Support: ഒരു വസ്തുവിനെ വിവിധ വശങ്ങളിൽ നിന്ന് നോക്കുമ്പോൾ ലഭിക്കുന്ന ചിത്രങ്ങൾ.</p>

</div>

<section class="concepts">

<h3>2.1 Theory of Projection</h3>

<div class="grid two">

<article class="card">

<h3>First Angle Projection</h3>

<p>Standard in India and Europe. The object is between the observer and the plane. Symbol: (First Angle Projection Symbol)</p>

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</article>
<article class="card">
  <h3>Third Angle Projection</h3>
  <p>Standard in USA. The plane is between the observer and the object. <strong>Top View (TV)</strong>
</article>
</div>

<h3>2.2 Projections of Points and Lines</h3>
<div class="topic-grid">
  <article class="topic-card">
    <h3>Points in 4 Quadrants</h3>
    <ul>
      <li><strong>1st Quadrant:</strong> Above HP, In front of VP (FV above XY, TV below).</li>
      <li><strong>2nd Quadrant:</strong> Above HP, Behind VP (Both above XY).</li>
      <li><strong>3rd Quadrant:</strong> Below HP, Behind VP (TV above XY, FV below).</li>
      <li><strong>4th Quadrant:</strong> Below HP, In front of VP (Both below XY).</li>
    </ul>
  </article>
  <article class="topic-card">
    <h3>Lines (1st Quadrant)</h3>
    <p>A line can be parallel to both planes, parallel to one and perpendicular to the other, <strong>
  </article>
</div>

<h3>2.3 Sectional Views</h3>
<p>Used to show internal features clearly. Imagine a <strong>Cutting Plane</strong> passing through
<div class="notice info">
  <strong>Hatching Rules:</strong> Use thin parallel lines at 45°. Hatching is only done on the
</div>
</section>

<section class="practical-centre">
  <h3>Lab Exercises (Module II)</h3>
  <article class="experiment" id="exp-ii-1">
    <header><span class="exp-no">Ex. II.1</span><div><h3>Orthographic Projection</h3></div></header>
    <div class="official-outcome"><strong>Outcome:</strong> Draw FV, TV, and Side View of a given
    <div class="exp-grid">
      <section><h4>Steps</h4><ol><li>Identify the direction of Front View (usually marked by an arrow)
      <section class="safety"><h4>Checklist</h4><ul><li>Are all visible edges thick?</li><li>Are
    </div>
  </article>
</section>
</section>

<section class="module" id="module-III">
  <div class="module-head">
    <div>
      <p class="eyebrow">Module III</p>
      <h2>CAD Basics</h2>
      <p class="lead">Transition from paper to screen. Learn the fundamental commands and settings
    </div>
    <div class="module-meta">
      <span>9 Hours</span><span>C03</span><span>Familiarize</span>
    </div>
  </div>

  <div class="grid two">
    <article class="card">
      <h3>Initial Setup</h3>
      <ul>
        <li><strong>UNITS:</strong> Set to Millimeters, Precision 0.00.</li>
        <li><strong>LIMITS:</strong> Set drawing area (e.g., 297,210 for A4). Use 'Zoom All'.</li>
        <li><strong>OSNAP:</strong> Enable Endpoint, Midpoint, Center, Intersection for precision.

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</article>
<article class="question-card">
  <span class="qtype">Practical ESE</span>
  <h3>Exam Marking Scheme (40 Marks)</h3>
  <table>
    <tr><td>10 Short Questions (Theory/Manual)</td><td>10 Marks</td></tr>
    <tr><td>CAD Drawing (Orthographic/Sectional)</td><td>18 Marks</td></tr>
    <tr><td>Viva Voce</td><td>8 Marks</td></tr>
    <tr><td>Record Book</td><td>4 Marks</td></tr>
  </table>
</article>
</div>
</section>

<footer class="footer">
  <p>&copy; 2026 POLY PMNA | Revision 2026 Educational Resource</p>
  <p>Course 1004: Engineering Drawing with CAD</p>
</footer>
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